XR CONCEPT PROPOSAL

VR Graffiti.

Learning the motion behind the mark

A virtual studio for guided lettering and art made from recorded gestures.

Smart Chen / Solo project / Concept animation



A mark on the wall

A gesture recorded through time

The hidden movement

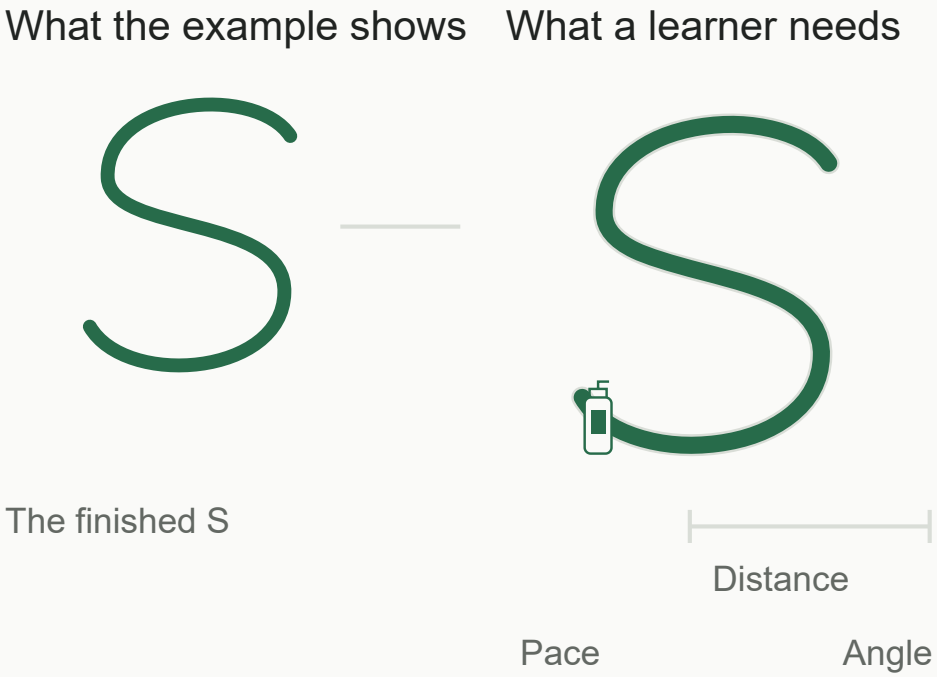
For beginners in lettering and street art, a finished letter hides the technique.

A place to practice

Permitted wall space, consumable paint, and suitable ventilation.

A process to learn

Distance, speed, angle and stroke order change the resulting mark.



18 studies in a review of observational motor learning in physical education

Evidence for demonstration, not proof for VR graffiti.

Han et al., 2022. Krylon, How to Spray Paint User needs are design assumptions to validate with beginners.

Existing tools and the learning gap

Approach	Established strengths	Focus of this proposal
Kingspray	VR spray, replay, reference projection and interactive tutorials*	A complete lesson with motion cues, fading support and unguided assessment
Open Brush	Expressive painting in 3D space	Reusing a learner's recorded gesture for practice and temporal composition
Video + physical practice	Artist demonstrations and real material feedback	Tracked tool motion with feedback relative to a shared virtual wall

The proposed contribution is an integrated learning sequence, with outcomes that still need testing.

Kingspray, official PC listing. Open Brush, official site *Kingspray features above refer to its official PC description. No claim of feature parity across versions.

Lettering is the learning task

Motor skills stay inside a meaningful creative exercise.



Tag / Handstyle

Rhythm, continuity and spacing

CURRICULUM DIRECTION



Bubble

Curves, smooth speed and stand-off distance

SEMESTER LESSON: S



Block

Straight strokes, turns and even coverage

CURRICULUM DIRECTION



Wildstyle

Layering and complex letter structure

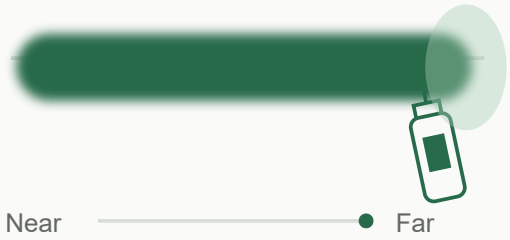
FUTURE SHOWCASE

In the S lesson: choose a tool, control a curve, reproduce it independently, then change its proportions and color deliberately.

Illustrative style families with artistic variation. These sketches do not define a universal graffiti alphabet. Concept animations.

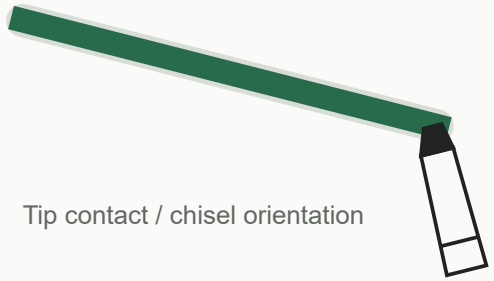
One studio, three ways to paint

Tool station | Learning wall | Free creation wall | Spatial canvas (stretch)



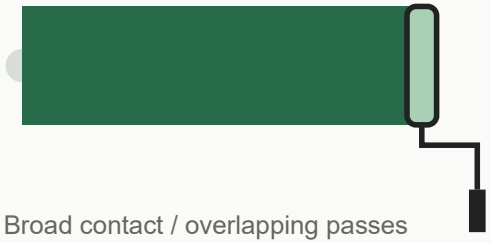
Spray can

Trigger to spray.
Distance changes the footprint.



Paint marker

Touch the wall with the tip.
Orientation shapes a crisp edge.



Paint roller

Move a broad contact area.
Practice coverage and overlap.

Controllers approximate tool behavior. Real pressure, friction and weight remain absent.

Concept animations. Spray is the priority. Marker and roller share a deliberately simple paint system.

A first lesson: Bubble S

The guide gradually gives control back to the learner.



Ghost tool + motion trail

Learner tool

Trace with a fading guide

01 **Observe**

Replay hand, tool pose and pace.

02 **Trace**

Follow the trail. Check distance.

03 **Reproduce**

Hide cues. Paint independently.

04 **Remix**

Change the letter with intention.

Proposed learner scenario and concept animation. Feedback can be requested again. Remix has no single correct trajectory.

What the guidance teaches



Target mark on wall 3D motion trail

The wall shows the mark.
The space in front shows the movement.

Observational learning

The ghost exposes stroke order and timing before the learner attempts the letter.

Scaffolding and fading

A full demonstration becomes a faint trail, then an attempt with no overlay.

Feedback and reflection

A distance cue during practice.
A trajectory comparison after the stroke.

Embodied interaction

Tracked tool motion controls the mark.
The learner practices relative to the wall.

Han et al., 2022. Wulf et al., 2010. Yu et al., CHI 2024. Levac et al., 2019 These are design choices to test, not guaranteed learning effects.

A pilot with an unguided test

Does ghost guidance help beginners acquire basic VR graffiti technique?

Optional exploratory pilot

A

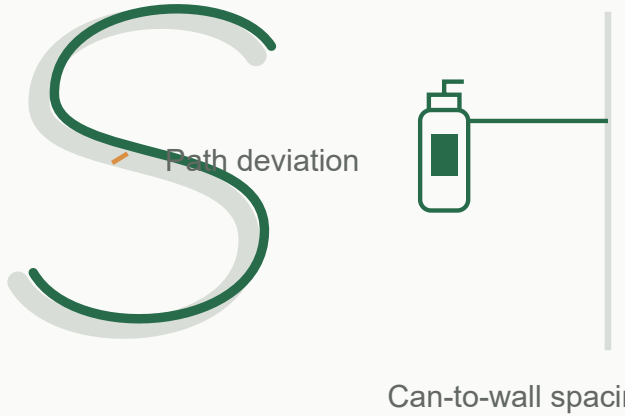
Conventional visual demonstration + target

B

Ghost hand + tool and spatial trail

Unguided pretest	Equal-duration practice	Unguided posttest

Random assignment to A or B. Same lesson and hardware.
Small pilot: descriptive findings and feasibility.
Delayed retention test if feasible.



Painted trajectory error
Mean nearest-target distance in the wall plane (mm)

Distance stability
Variation of tool-to-wall distance while spraying (cm)

Compare pre/post change in each metric. Physical transfer needs a separate real-world test.

Levac et al., 2019 Proposed study only. No participant results. Example geometry. Participant work follows course consent requirements.

The recording becomes material

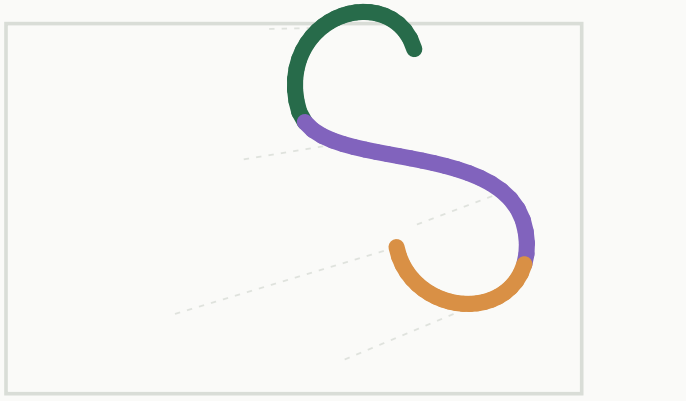
One gesture contains a history, not just a finished image.



Earlier attempts — Now

Temporal Layers

Earlier attempts remain visible
as translucent layers of the artwork.



Earlier / near wall — Later / deeper

Time-to-Space

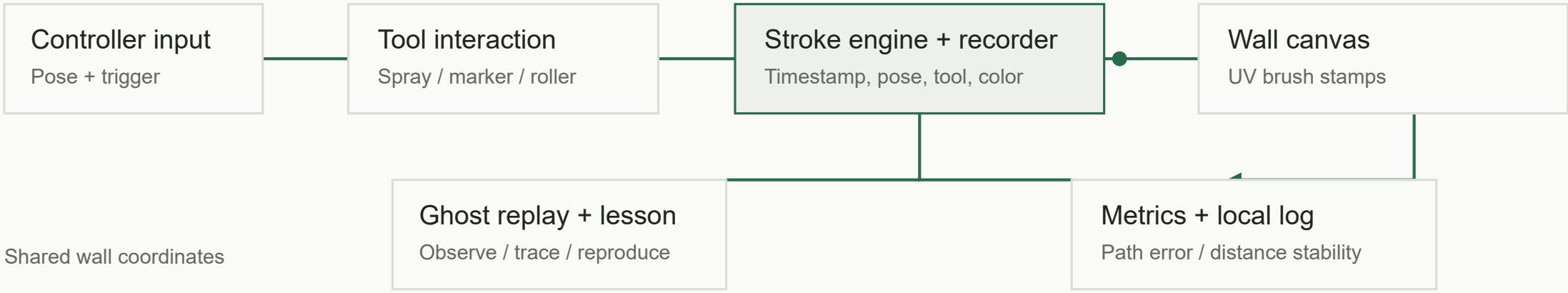
Elapsed time maps to depth,
turning a flat stroke into a spatial line.

Creative research beyond the semester MVP. Both reuse timestamped pose and stroke data.

Concept animations. Spatial painting and replay already exist. The proposed focus connects learning with temporal reinterpretation.

A shared stroke system

Meta Quest + controllers / Unity + C# / OpenXR + XR Interaction Toolkit



Paint and GPU cost

Bounded RenderTexture + brush stamps. Profile on Quest. Reduce resolution and cap stroke history if needed.

Replay alignment

Store poses in wall coordinates. Calibrate wall height and reach. Keep the first lesson on one plane.

Tool fidelity and scope

Use visual distance cues. Prioritize spray and one complete lesson. Simplify contact tools before expansion.

Unity, XR input options Proposed architecture. Local recording requires no backend. No fluid simulation or fast hand-tracking dependency.

A semester-sized commitment

Smart Chen, solo developer Unity XR, interactive systems, visual design and media production.
I own interaction design, implementation, visual direction, logging and evaluation planning.

PHASE 1

Foundations

Quest input + studio.
Spray draws on the wall.

PHASE 2

Tools + lesson

Basic marker and roller.
Bubble S target + lesson UI.

PHASE 3

Guided practice

Record + replay.
Ghost trail + fading cues.

PHASE 4

Test + polish

Unguided attempt + logs.
Usability checks + stable demo.

Semester MVP

One studio. Three basic tools. One complete S lesson.
Ghost guidance, independent reproduction and basic remix.

If the core is stable

Small pilot, replay scrubbing,
Temporal Layers and Time-to-Space.

First milestone: a responsive stroke on the wall.

Relative milestone sequence. Calendar dates depend on the course’s final deadline. Concept stage: no prototype or study results yet.

References

Research informs the design. It does not establish outcomes for this application.

Han, Y., Ali, S. K. B. S., & Ji, L. (2022)

Use of Observational Learning to Promote Motor Skill Learning in Physical Education: A Systematic Review.
IJERPH, 19(16), 10109. 18 included studies.

Levac, D. E., Huber, M. E., & Sternad, D. (2019)

Learning and transfer of complex motor skills in virtual reality: a perspective review.
Journal of NeuroEngineering and Rehabilitation, 16, 121.

Yu, X., Lee, B., & Sedlmair, M. (2024)

Design Space of Visual Feedforward And Corrective Feedback in XR-Based Motion Guidance Systems.
CHI 2024. DOI: 10.1145/3613904.3642143.

Wulf, G., Shea, C., & Lewthwaite, R. (2010)

Motor skill learning and performance: a review of influential factors.
Medical Education, 44(1), 75–84.

Infectious Ape. Kingspray Graffiti VR

Official PC product description on Steam.
Spray, replay, projector and interactive tutorial features.
Features can differ across platform versions.

Open Brush

Official project site.
Room-scale painting in three-dimensional space.

Krylon. How to Spray Paint

Manufacturer guidance on technique and suitable ventilation.
Used for material-practice context.

Unity. XR input options

Official Unity 6 documentation.
OpenXR, controller tracking and XR Interaction Toolkit.

Linked titles open original sources. Product pages and documentation checked October 27, 2026. Backup slide, outside the 7:30 talk.

Discussion

Questions to guide the next iteration

Why VR instead of a video?

Show a tracked tool's motion relative to the wall, then let the learner practice in that same space.

Why keep three tools?

They share mark and recording infrastructure. Spray gets priority. Contact tools stay basic.

How will remix be assessed?

Use reflection or artist feedback.
Trajectory error measures copying, not creativity.

Will this transfer to real paint?

Unknown. Controllers lack material feedback. A physical task would require a separate test.

What if the plan is too large?

Protect one spray lesson and useful logging. Defer the study and temporal art interactions.

What is the first thing to validate?

Can a beginner understand the ghost and trail without losing sight of their own mark?

Backup for the 3–5 minute Q&A. Physical transfer and comparative learning benefits remain open research questions.